

ANIRUDH RAGHAVAN

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EDUCATION

Purdue University, West Lafayette, IN – MS in Autonomy (GPA 3.83/4.0)

Aug 2024 – May 2026

Vellore Institute of Technology, Chennai, India – B. Tech, Electronics and Communication Engineering (GPA 8.11/10) 2020 – 2024

TECHNICAL SKILLS

Languages & Frameworks: Python, C++, PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Scikit-learn

Robotics & Embedded: ROS2, Gazebo, RViz, NVIDIA Jetson Nano

Computer Vision & Machine Learning: Object Detection (YOLO), Semantic & Video Object Segmentation (SegFormer), CNNs, Transformers, Model Fine-Tuning, Transfer Learning, Optical Flow (RAFT), Feature Fusion, Image Processing

Tools & Platforms: Git, Linux (Ubuntu)

EXPERIENCE

Data Science Fellow – Chicago Education Advocacy Coalition (ChEAC)

Aug 2026 – Present

Remote, USA

- Proposed and scoped an AI-powered inventory assistant for a community food bank, designing a computer vision and OCR pipeline to digitize donated-item photos and a RAG-based conversational search so volunteers and community members can find food by allergen, diet, or expiration date in plain language.

Graduate Research Project (Faculty-Advised) – Purdue University

Aug 2025 – Dec 2025

West Lafayette, IN

- Built an end-to-end Python benchmarking framework evaluating YOLOv8, YOLOv9, YOLO12 and Autoware's YOLOX detector, converting Waymo Open Dataset 3D labels into standardized 2D bounding boxes across a unified training and evaluation pipeline.
- Fine-tuned YOLOv8 to 65% mAP@0.5 overall and 85% mAP@0.5 with 86.8% recall on pedestrians.
- Outperformed Autoware's production YOLOX baseline (39% mAP@0.5) by more than 2x on the safety-critical Person category.

SELECTED PROJECTS

AI-Enabled Earthquake Rescue System (published research) | Python, YOLOv8, NVIDIA Jetson Nano, Transfer Learning | *Spring 2024*

- Fine-tuned YOLOv8 via transfer learning on 6,837 images to detect trapped victims under rubble, achieving 93.1% mAP@0.5, 90.5% recall, and 90% F1-score overall, with 98.2% precision in face detection and 94.6% in palm detection.
- Deployed the model on an NVIDIA Jetson Nano integrated with a mobile robotic vehicle and camera, enabling near real-time on-device inference and cloud-based alert transmission to rescue teams.

Structured Slot Aggregation for Cross-Modal Video Object Segmentation | PyTorch, Transformers, SegFormer | *Fall 2025* | [GitHub](#)

- Designed a dual-stream segmentation framework using guided slot attention to maintain consistent object-centric representations across video frames.
- Resolved temporal instability in high-motion scenes by fusing RAFT-based optical flow with SegFormer visual features, keeping temporal mask drift below 5% under occlusion and rapid camera motion.

Weather-Invariant Object Detection | Python, YOLOv8, OpenCV, Data Augmentation | *Fall 2024* | [GitHub](#)

- Developed a two-stage adaptive perception pipeline that integrates a CNN-based weather classifier to trigger specific image preprocessing (e.g., dehazing, gamma correction) before inference.
- Prioritized edge-computing efficiency by avoiding full model retraining and improved robustness in adverse conditions, resulting in a 27% reduction in false negatives and 93.1% mAP@0.5 (82.3% mAP@0.5:0.95), 85.5% precision and 87.4% recall on custom test splits.

PUBLICATIONS

- Anirudh Raghavan**, Y. Ramesh, H. Shyam, B. Hency V., M. N. Halgamuge - "Enhancing Earthquake Response with AI: Real-Time Human Detection Using Deep Learning and Robotic Systems," *Results in Engineering (Elsevier, Q1, Impact Factor 7.9)*, 2026. | [Link](#)
- Abhishek Sebastian, R. Pragna, K. Vishal Vythianathan, Dasaraju Sohan Sai, U. Shiva Sri Hari AI, **R. Anirudh**, Apurv Choudhary - "Design of Rubble Analyzer Probe Using ML for Earthquake," *AIP Conference Proceedings 2946(1): 040003*, 2023. | [Link](#)